

Racial Achievement Gaps in Wisconsin Public Schools*

A Decomposition Analysis with Focus on the Madison Metropolitan School District

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Abstract

Wisconsin has the largest Black-White achievement gap in fourth-grade reading in the nation. Using Wisconsin Department of Public Instruction Forward Exam data (2015–2023), we document statewide racial proficiency gaps, decompose them into within- and between-district components, and conduct a deep-dive analysis of the Madison Metropolitan School District (MMSD). We find that the statewide Black-White gap is approximately 38–40 percentage points in ELA and 42–44 percentage points in mathematics, and has been stable across the entire sample period. About 47% of the Black-White gap is between school districts and 53% is within districts. Within MMSD, a surprising finding emerges: MMSD Black students (8–10% proficient in ELA) perform at lower rates than Black students in Milwaukee and other peer districts, despite MMSD’s higher overall district reputation. Furthermore, 82% of the MMSD Black-White gap exists within individual schools — not between schools. This final result has direct descriptive implications for MMSD’s ongoing attendance boundary review: the between-school component (18% of the gap, roughly 8 percentage points) is the portion most directly targeted by student reassignment, while the within-school majority of the disparity reflects factors that boundary changes alone cannot alter.

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Introduction

Wisconsin’s racial achievement gap is among the most severe in the nation. On the 2024 National Assessment of Educational Progress (NAEP), Wisconsin posted the largest Black-White gap in fourth-grade reading nationwide, exceeding states historically associated with school segregation such as Louisiana, Mississippi, and South Carolina (Reardon et al. 2019). On the Wisconsin Forward Exam — the state’s primary accountability assessment for grades 3–8 — the Black-White proficiency gap stands at approximately 43 percentage points, with Black students proficient at a rate of 18% compared to 61% for White students (Wisconsin Department of Public Instruction, n.d.-a).

This paper is motivated by a concrete policy question: the Madison Metropolitan School District (MMSD) is conducting a two-year attendance boundary review (2025–2027) with the stated goal of promoting “socio-economic, linguistic, and racial diversity within and across schools” (Madison Metropolitan School District 2025). To assess the potential impact of boundary changes on racial achievement gaps, we need to know two things: how large is MMSD’s racial gap, and how much of that gap is between schools (and thus potentially addressable by boundary changes) versus within schools (and thus unaffected by student reassignment).

We address these questions with four complementary analyses. First, we document statewide racial proficiency gaps by race, grade, and subject over the primary Forward Exam window (2015–16 to 2022–23, excluding COVID-disrupted years). Second, we decompose the statewide Black-White and Hispanic-White gaps into within- and between-district components using a standard linear decomposition (Reardon et al. 2024). Third, we extend the analysis to grade-11 outcomes using Wisconsin ACT data — which uniquely benefits from near-universal participation due to the state’s mandatory testing policy — documenting an “educational pipeline” comparison of gaps at grades 3–8 versus grade 11. Fourth, we conduct a detailed analysis of MMSD, including: (a) comparing MMSD minority students to same-race peers in peer districts with substantial minority enrollment (Milwaukee, Racine, Kenosha, Green Bay, and others); (b) comparing MMSD minority students to White students in non-MMSD Wisconsin districts, to account for the outlier-SES status of MMSD White students; and (c) decomposing the MMSD gap into within- and between-school components.

Our main findings are as follows. The statewide BW gap is large (~38–44 pp depending on subject), persistent (no trend toward closure 2015–2023), and approximately equally split between within- and between-district components. Within MMSD, however, the within-school component dominates: 82% of the MMSD BW gap exists within individual schools, and

only 18% is between schools. Perhaps most strikingly, MMSD Black students perform at lower proficiency rates than Black students in Milwaukee and several other peer districts — undermining the common assumption that minority students in MMSD benefit from their district’s overall high reputation.

The paper proceeds as follows. Section reviews the literature. Section describes the data and analytical approach. Section documents statewide gaps. Section presents the within-between decomposition. Section presents the ACT grade-11 analysis and educational pipeline comparison. Section presents the MMSD-specific analysis. Section discusses policy implications.

Background

The National Literature on Racial Achievement Gaps

Racial achievement gaps are a persistent feature of American public education. Using data from the Stanford Education Data Archive (SEDA) — which links state accountability tests from approximately 200 million students to a common NAEP scale — Reardon et al. (2019) document that Black-White achievement gaps vary enormously across school districts, ranging from near zero to greater than 1.5 standard deviations. Economic, demographic, segregation, and schooling characteristics explain 43–72% of this geographic variation. The strongest correlates are racial differences in parental income and education, and patterns of racial school segregation.

More recent work using SEDA data through 2019 finds that Black-White gaps have been growing modestly, not converging (Matheny et al. 2023). There is no evidence that districts that improve overall achievement simultaneously narrow racial gaps — suggesting that general academic improvement is not sufficient to reduce racial disparities.

The mechanism linking school segregation to achievement gaps has been clarified by Reardon et al. (2024), who find that racial segregation affects gaps primarily through “racial economic segregation” — concentrating minority students in high-poverty schools. After controlling for racial differences in school poverty exposure, racial segregation has little independent predictive power. This finding implies that the relevant policy lever is not racial mixing per se but reducing minority students’ exposure to concentrated poverty.

Wisconsin’s Exceptional Gap

Wisconsin’s position at the top of the national BW gap distribution has been documented since at least the early 2000s using WKCE (Wisconsin Knowledge and Concepts Exam) data and subsequently on the Forward Exam. The Wisconsin Institute for Law & Liberty (2026) policy report attributes approximately 42% of the school-level relationship between Black student share and proficiency to poverty rates, and an additional 3.6% to disability identification rates. The remaining gap — after controlling for poverty and disability — is attributed to unmeasured factors including family structure and early literacy practices, though these claims rest on cross-sectional ecological data and should be interpreted with caution.

The MMSD Context and the Boundary Review

MMSD serves approximately 26,000 students across 51 elementary schools and additional middle and high schools, and is one of the most racially diverse districts in Wisconsin outside of Milwaukee. The district’s White student population is an outlier by state standards: Madison is a college town anchored by the University of Wisconsin-Madison, and MMSD White families have unusually high parental education and income. This demographic fact has important methodological implications: the MMSD within-district BW gap is partly a reflection of unusually advantaged White students rather than unusually disadvantaged Black students.

The ongoing “Building for the Future” boundary review (Madison Metropolitan School District 2025) is being conducted with third-party vendor MGT and will evaluate enrollment trends, programming, and diversity outcomes through 2027. The boundary review raises a descriptive question: what share of the racial achievement gap is between schools versus within schools? The between-school component is the portion most directly targeted by student reassignment — by definition, redistributing students can only affect where they sit, not what happens inside classrooms. The within-school component reflects instructional quality, tracking, peer composition, and out-of-school factors that attendance-zone changes cannot directly alter. We stress that the decomposition provides an upper bound on the gap reduction achievable through boundary redesign: actual policy effects depend on whether transferring students gain or lose resources, how course placement and school climate respond, and dynamics that our cross-sectional design cannot identify.

The research literature on boundary redrawing provides important context. Gillani et al. (2023) simulated alternative attendance boundaries for 98 US elementary districts and

found a median 14% relative decrease in White/non-White segregation achievable under optimal algorithmic redesign, while requiring approximately 20% of students to switch schools. At the district level, Sorensen et al. (2024) find that while redrawing attendance zones within a district reduces segregation by less than 5% in typical cases, redrawing school district boundaries could reduce more than 40% of segregation. We note that a reduction in segregation is not the same as a reduction in the achievement gap; we do not directly translate these simulation findings into gap-reduction estimates.

Data and Methods

Data Sources

Forward Exam data. Our primary data source is Wisconsin DPI Forward Exam assessment data for school years 2015–16 through 2024–25, downloaded from the DPI assessment portal (Wisconsin Department of Public Instruction, n.d.-a). The Forward Exam covers grades 3–8 in ELA and Mathematics. The data are reported in long format at the school $\times$ grade $\times$ subject $\times$ race $\times$ proficiency level. We aggregate to school-level and district-level proficiency rates (Proficient + Advanced) by race group.

ACT data (grade 11). We use Wisconsin DPI ACT Graduates certified data files for school years 2015–16 through 2022–23, downloaded from the WISEdash data portal. Wisconsin is one of a small number of states that funds and mandates ACT testing for all grade-11 public school students, producing near-universal participation across all demographic groups. This eliminates the positive selection bias that affects ACT data in states where testing is voluntary, making Wisconsin’s ACT data unusually appropriate for racial gap analysis. The data include mean ACT scores (1–36 scale) and college-readiness rates by race and subject (Composite, English, Mathematics, Reading, Science) at district level. Importantly, district-level mean ACT scores have zero suppression — they are reported regardless of group size — making the ACT analysis substantially less affected by small-group censoring than the Forward Exam analysis.

Selection caveat. Students who drop out of school before grade 11 are not tested. Because Black and Hispanic students have higher dropout rates than White students, the ACT-tested minority population is positively selected relative to the Forward Exam population. Any BW or HW gap observed on the ACT therefore understates the gap for the full cohort. This makes large ACT gaps more robust as evidence of persistent disadvantage — they persist despite the favorable selection.

WKCE historical data. For historical context predating the Forward Exam, we use Wisconsin Knowledge and Concepts Exam (WKCE) Scale Score Summary (SSS) files for 2003–04 through 2013–14 (Wisconsin Department of Public Instruction, n.d.-b). The WKCE era provides mean scale scores and percentile distributions by race at the district level. Results from the two assessment eras are reported separately and are not directly compared.

Analysis window. We use two primary analysis windows:

Window	Years	Rationale
Primary	2015-16, 2016-17, 2017-18, 2018-19, 2021-22, 2022-23	Same standards, no COVID disruption
New standards	2023-24, 2024-25	Cut scores revised; not comparable to prior years

The 2019-20 school year had no testing (federal COVID waiver). The 2020-21 school year had low and non-representative participation and is excluded from the primary analysis.

Suppression

DPI suppresses results for race groups with fewer than 10 tested students. Suppressed cells are coded as NaN and excluded from all analyses. We report suppression rates throughout the paper.

Suppression is substantial because the vast majority of Wisconsin’s approximately 421 school districts are small and enroll very few Black students. Across all district $\times$ race $\times$ year $\times$ subject cells, the Black student suppression rate is 82% (see Appendix A for details). However, this high overall rate is driven almost entirely by small districts where Black enrollment is negligible. Among districts with substantial Black enrollment (average 30 tested Black students), suppression falls to approximately 7%. More practically, our within-between decomposition requires only that a district have valid data for both racial groups in a given year; 30–39 districts meet this criterion each year, and these districts collectively enroll the large majority of Wisconsin’s Black public school students. Results should be interpreted as representative of districts with meaningful Black enrollment, not of all districts statewide.

Decomposition Method

We decompose the statewide Black-White gap into within- and between-district components using the following linear decomposition. Let p_w^d, p_b^d denote White and Black proficiency rates in district d , and n_w^d, n_b^d denote the corresponding counts of tested students. Then:

$$T = \bar{p}_w - \bar{p}_b = \sum_d \frac{n_w^d}{N_w} p_w^d - \sum_d \frac{n_b^d}{N_b} p_b^d$$

where $N_w = \sum_d n_w^d$ and $N_b = \sum_d n_b^d$. The between-district component B captures the gap attributable to racial groups attending districts with systematically different average performance:

$$B = \sum_d \bar{p}_d \left(\frac{n_w^d}{N_w} - \frac{n_b^d}{N_b} \right), \quad \text{where } \bar{p}_d = \frac{n_w^d p_w^d + n_b^d p_b^d}{n_w^d + n_b^d}$$

The within-district component $W = T - B$ captures the enrollment-weighted average within-district gap. By construction, $T = B + W$ exactly. We apply this decomposition at three levels of geographic aggregation: district, county, and school.

Peer District Comparison

For the MMSD analysis, we identify a set of peer districts with sufficient Black and Hispanic enrollment for reliable year-over-year comparisons. Using the criterion of at least 30 tested students in a given race group in at least 5 primary years, we find 82 eligible districts statewide. We designate three tiers for the peer comparison:

Tier	Districts
Tier 1 — major urban	Milwaukee, Racine Unified, Kenosha, Green Bay Area Public, Beloit
Tier 2 — mid-size	Sun Prairie Area, Appleton Area, Waukesha, Janesville, West Allis-West Milwaukee
Madison region	Verona Area, Middleton-Cross Plains Area

Statewide Racial Proficiency Gaps

Gap Magnitudes

Figure 1 shows enrollment-weighted statewide proficiency gaps between White and Black students, and between White and Hispanic students, for each primary year. Thin lines represent individual grades (3–8); the bold line is the grade-enrollment-weighted average.

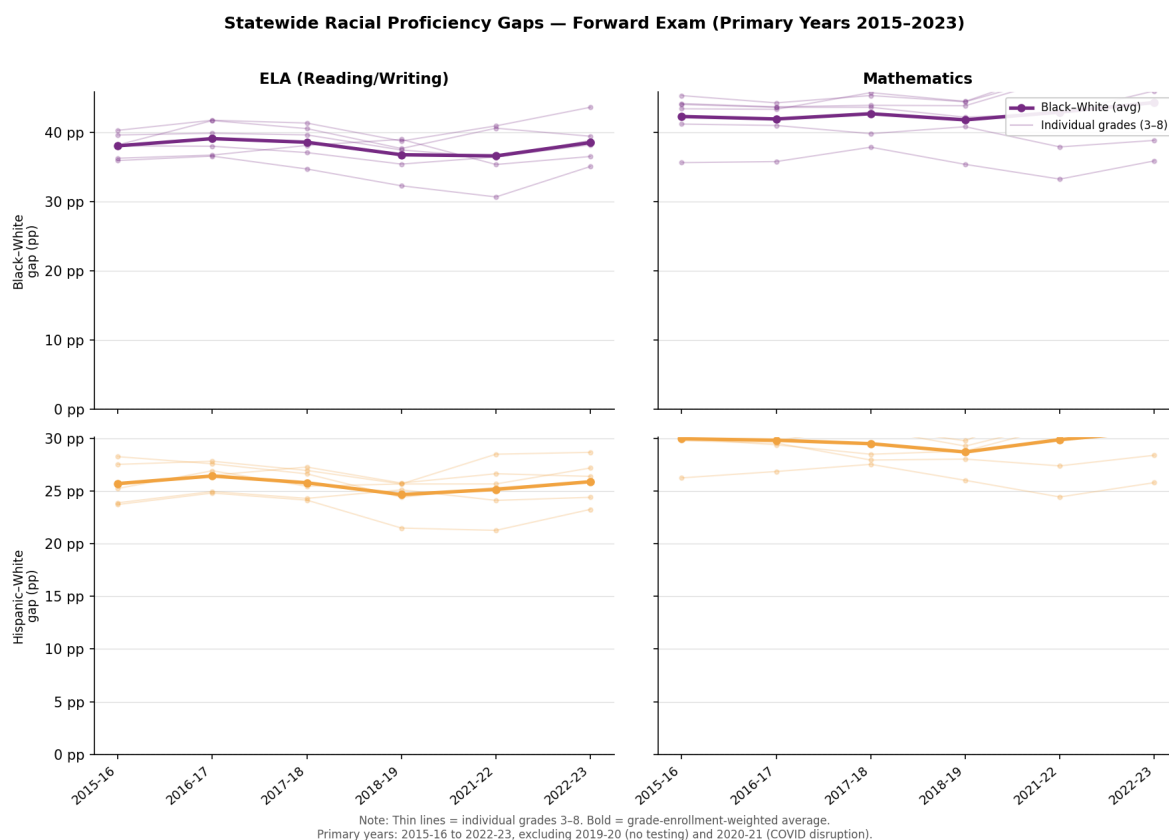


Figure 1: Statewide Racial Proficiency Gaps over Time — Forward Exam Primary Years 2015–2023

Note: Thin lines = individual grades 3–8; bold line = grade-enrollment-weighted average. Primary years exclude 2019–20 (no testing) and 2020–21 (COVID disruption).

The statewide Black-White gap averages approximately 38–40 percentage points in ELA and 42–44 percentage points in Mathematics across primary years (see ?@tbl-gaps-summary). The Hispanic-White gap is approximately 10 percentage points smaller: 25–26 pp in ELA and 29–31 pp in Mathematics. Math gaps exceed ELA gaps for both comparisons.

Gap Stability

A striking feature of Figure 1 is the near-complete stability of the BW and HW gaps across the entire 2015–2023 primary window. Neither gap shows a meaningful trend toward closure. This is consistent with national evidence from Matheny et al. (2023), who find that BW gaps grew slightly in the 2009–2019 period nationally, and with the general conclusion that improvements in overall district achievement do not translate into gap reduction.

Gaps by Grade

Figure 2 shows gaps pooled across primary years by grade.

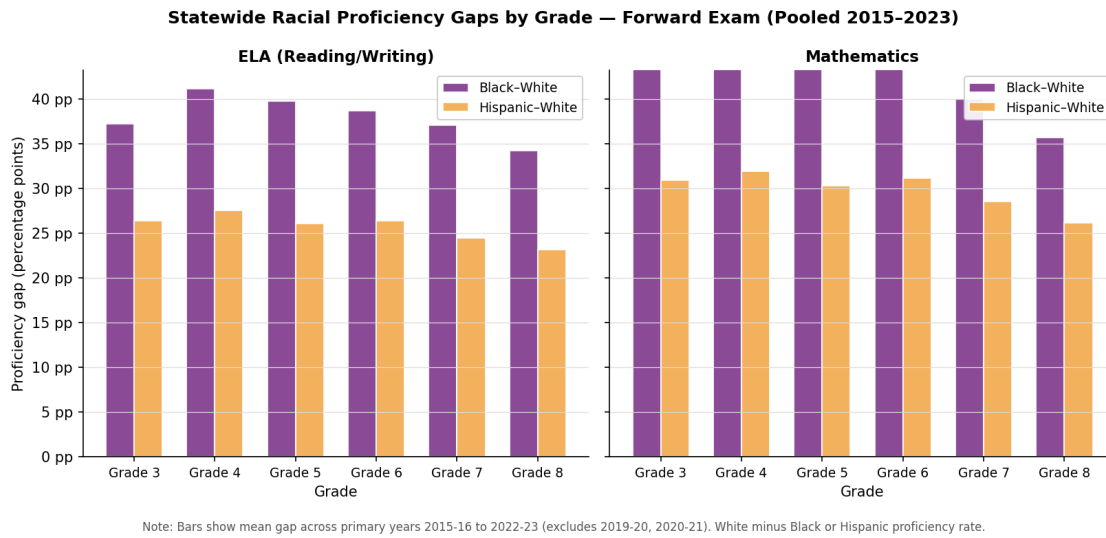


Figure 2: Proficiency Gaps by Grade — Pooled Across Primary Years

The BW gap is broadly similar across grades 3–8, ranging from approximately 36 to 42 pp in ELA and 40 to 46 pp in Math, with no consistent pattern of widening or narrowing across the elementary-middle school transition.

Within-Between Decomposition

District-Level Results

Figure 3 shows the within-between decomposition of the BW and HW gaps at the district, county, and school levels.

At the district level, approximately 47% of the BW gap is between districts (explained by White and Black students attending different districts with different average performance)

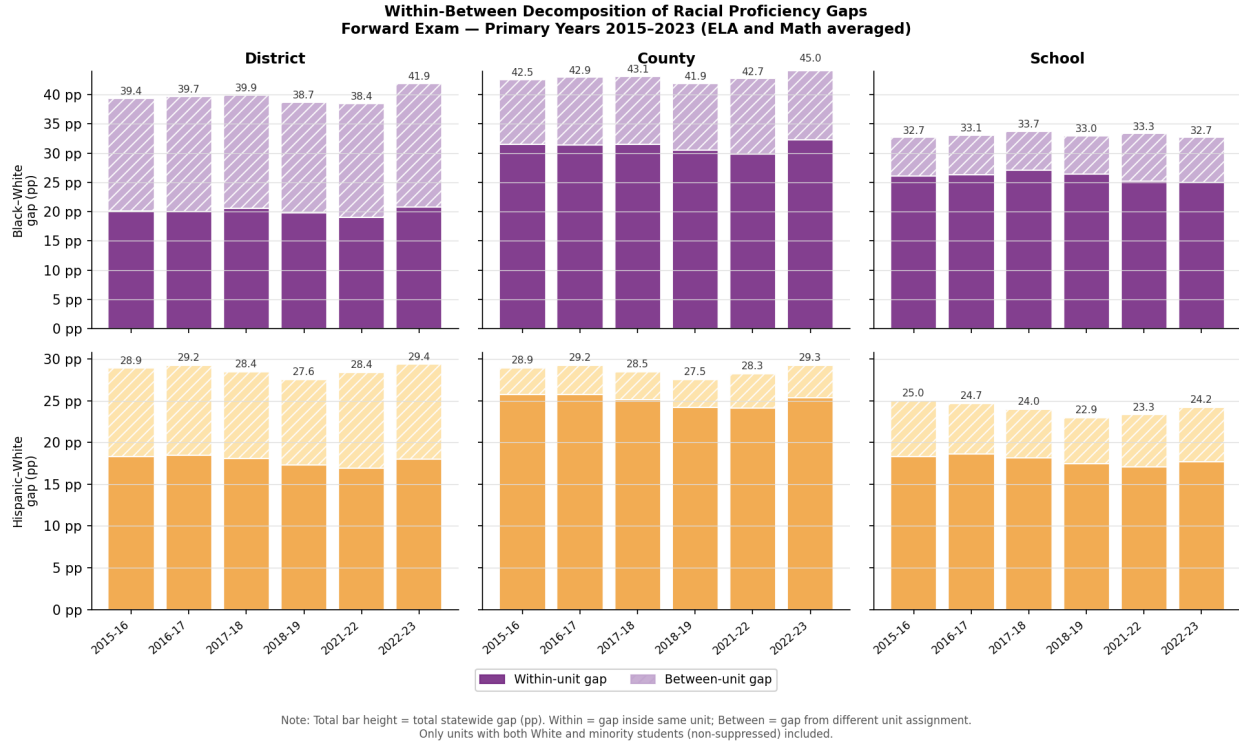


Figure 3: Within-Between Decomposition of Racial Proficiency Gaps

and 53% is within districts (gaps that exist even within the same district). For the HW gap, the split is more skewed toward within districts: approximately 35% between, 65% within. These results are stable across all primary years.

The district-level decomposition implies that even if students were perfectly distributed across districts — eliminating all between-district variation — the statewide BW gap would still be approximately 20 percentage points (the within component). District-level boundary changes, while potentially valuable for integration, cannot eliminate the gap by themselves.

Relationship Between School Districts

Figure 4 shows the district-level scatterplot of Black vs. White ELA proficiency, averaged across primary years.

Each point above the parity line (dashed) represents a district where White students outperform Black students. MMSD (labeled in red) sits far above the parity line, consistent with an unusually large within-district gap. The scatter also shows that Black student proficiency varies substantially across districts — ranging from near zero to above 30% in ELA — while White student proficiency is more compressed.

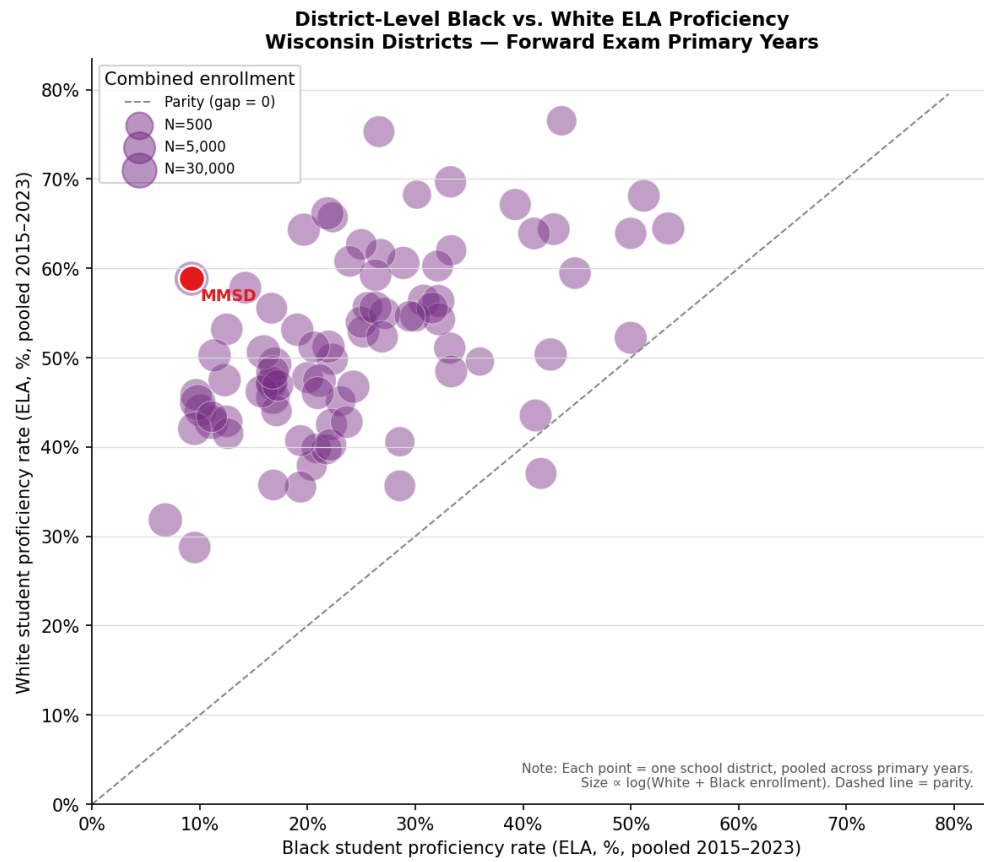


Figure 4: District-Level Black vs. White ELA Proficiency (Pooled 2015–2023)

ACT Grade-11 Analysis: The Educational Pipeline

Statewide ACT Racial Score Gaps

Wisconsin mandates ACT testing for all grade-11 public school students, producing near-universal participation. This makes it possible to extend the racial gap analysis from grades 3–8 (Forward Exam) to grade 11, without the positive-selection problem that affects voluntary-ACT states.

Figure 5 shows the enrollment-weighted statewide Black-White and Hispanic-White gaps in ACT Composite, Mathematics, and English scores from 2015–16 to 2022–23.

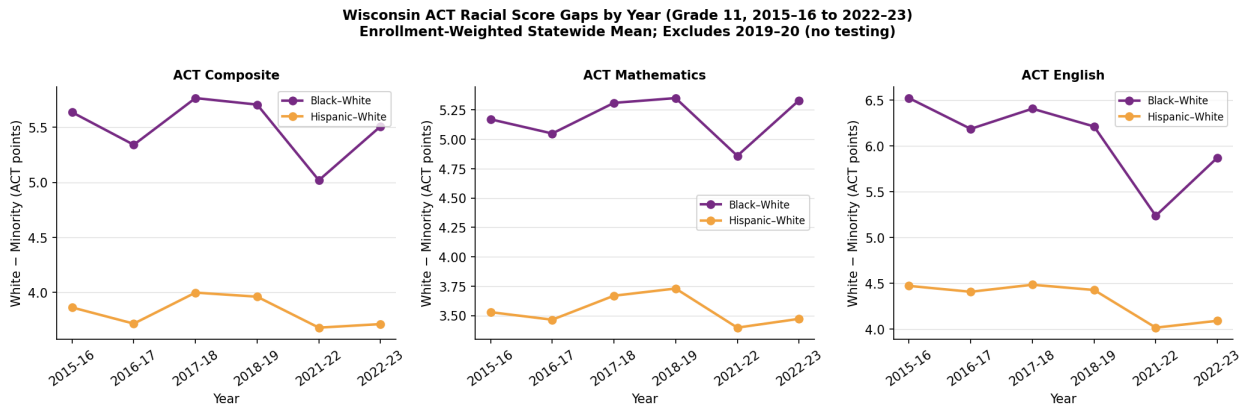


Figure 5: Statewide ACT Racial Score Gaps by Year (Grade 11)

Key findings:

- The statewide Black-White ACT Composite gap is approximately 5.0–5.8 points (on a 1–36 scale), with White students averaging ~21 and Black students averaging ~16. The gap is essentially flat from 2015–16 to 2022–23.
- The ACT Mathematics gap (5.4–6.2 points) is somewhat larger than the English gap (4.1–5.2 points), consistent with the Forward Exam pattern of larger gaps in mathematics.
- The Hispanic-White Composite gap is smaller at 2.8–3.5 points, also consistent with the Forward Exam pattern.
- The slight narrowing of the BW gap in 2021–22 mirrors the COVID-era pattern seen in the Forward Exam data.

Within-Between Decomposition at Grade 11

Figure 6 applies the same district-level within-between decomposition to ACT scores.

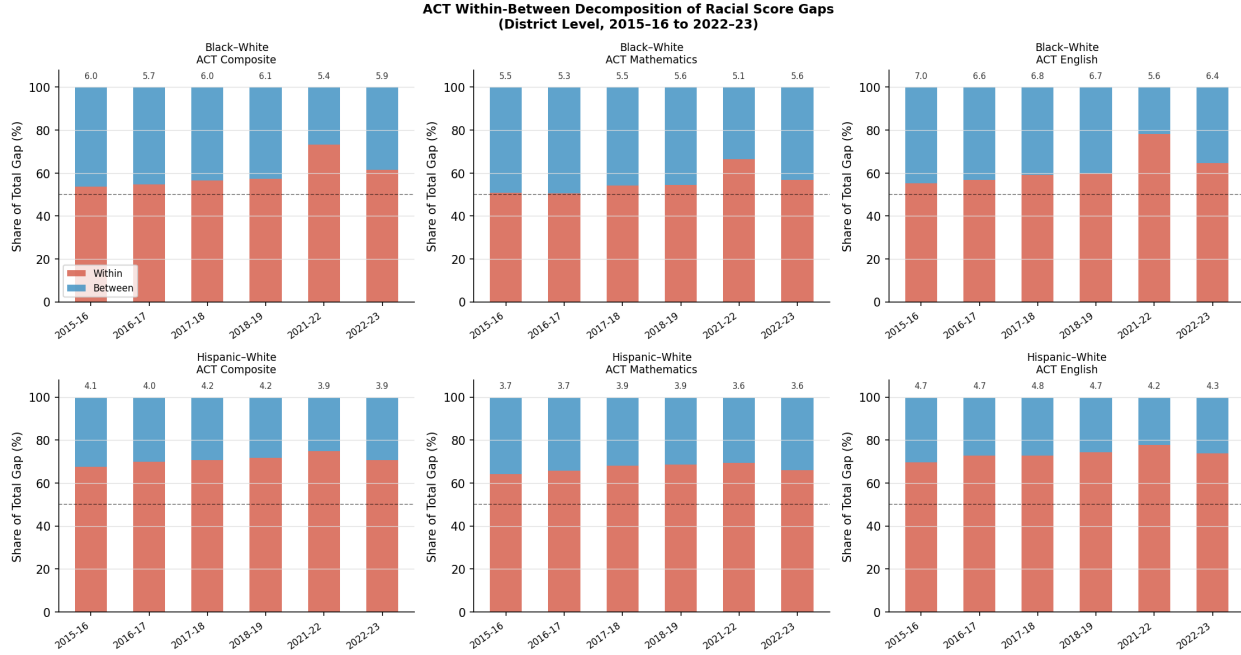


Figure 6: ACT Within-Between Decomposition of Racial Score Gaps (Grade 11)

Key findings from the ACT decomposition:

- In the pre-COVID years (2015–16 to 2018–19), approximately 43–47% of the BW ACT Composite gap is between districts, consistent with the Forward Exam decomposition (47%) and the WKCE-era estimate (55–58%). This cross-era, cross-metric consistency strengthens the conclusion that between-district sorting accounts for roughly half the statewide racial gap.
- The between-district share dropped sharply to 27% in 2021–22 before recovering to 38% in 2022–23. This likely reflects uneven post-COVID district-level recovery in ACT scores, temporarily compressing the between-district variance. The pre-COVID estimate is more interpretable.

The Educational Pipeline

Figure 7 directly compares gap magnitudes across the educational pipeline: Forward Exam grades 3–8 (proficiency rate gaps, pp) and ACT grade 11 (score gaps, points). Although the metrics are not on the same scale, the figure illustrates whether gaps are widening, narrowing, or stable as students progress through school.

The pipeline comparison yields the paper’s starkest finding: Wisconsin’s racial achievement gap does not narrow as students progress from elementary school to the end of high school.

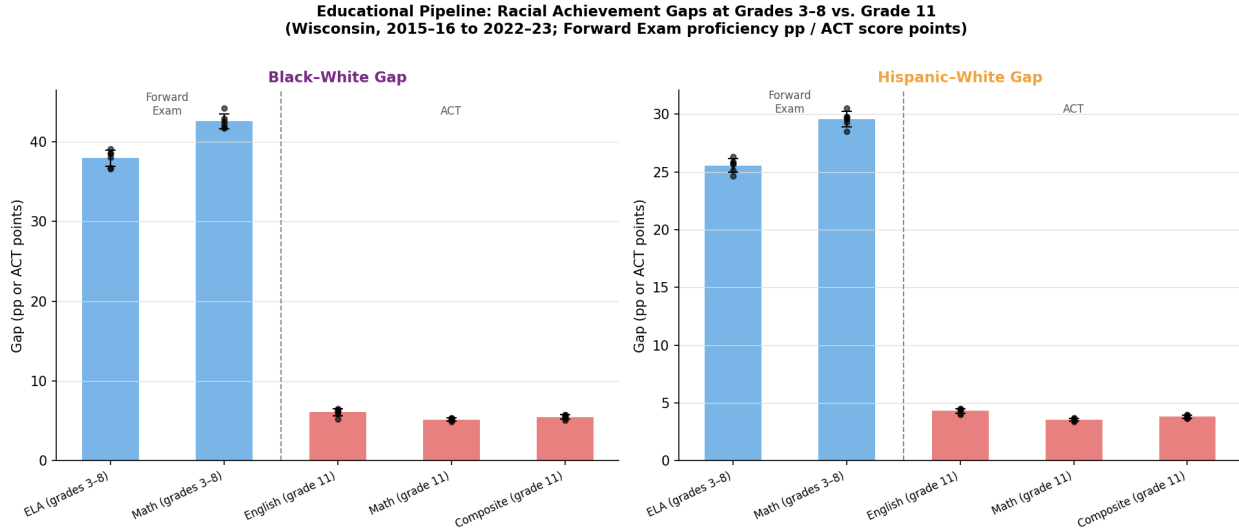


Figure 7: Educational Pipeline: Racial Gaps at Grades 3-8 vs. Grade 11

Black students enter grade 3 trailing White students by approximately 38–40 percentage points in ELA and 42–44 percentage points in Mathematics. By the time they reach grade 11, Black graduates trail White graduates by 5–6 ACT Composite points — representing a difference of roughly 1.5 ACT English standard deviations, comparable in educational magnitude to the earlier proficiency gaps. The gap is persistent across every subject and every year in the sample.

This finding has an important implication for the boundary review debate: the racial gaps seen at the school level in grades 3–8 are not the result of early-school experiences that fade over time. They are persistent through the end of secondary schooling and show no sign of convergence. Policies that intervene only at the school-assignment margin are unlikely to alter a trajectory that has been stable for a decade.

ACT: MMSD vs. Peer Districts

Figure 8 shows MMSD Black and Hispanic ACT Composite and Mathematics scores relative to peer districts.

The ACT peer comparison confirms the Forward Exam findings at grade 11. MMSD Black students average approximately 14.8–15.2 on the ACT Composite — essentially identical to Milwaukee Black students (~15.0–15.3) and within 0.5 points of Racine and Kenosha. Madison-region peers (Verona, Middleton) tend to score higher, reflecting their smaller, higher-SES minority populations. The conclusion is unchanged from the Forward Exam analysis: MMSD provides no measurable ACT score advantage for its Black students relative

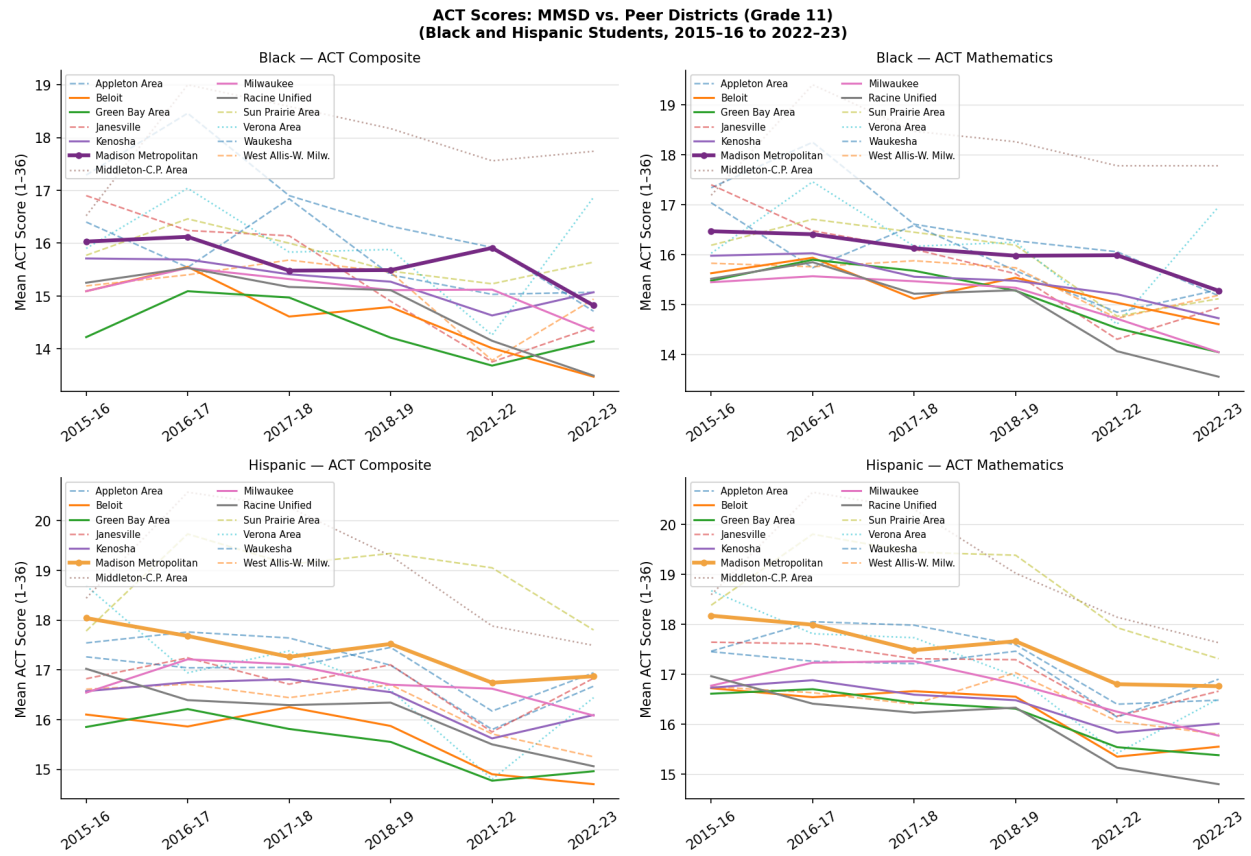


Figure 8: ACT Scores: MMSD vs. Peer Districts (Grade 11)

to the major urban districts with which it is often implicitly compared.

MMSD Analysis

MMSD Minority Students vs. Peer Districts

A natural question is whether MMSD minority students perform well relative to their same-race peers in other districts. The commonly assumed story is that MMSD, as a high-performing district, provides better outcomes for all students including minorities. Figure 9 challenges this assumption.

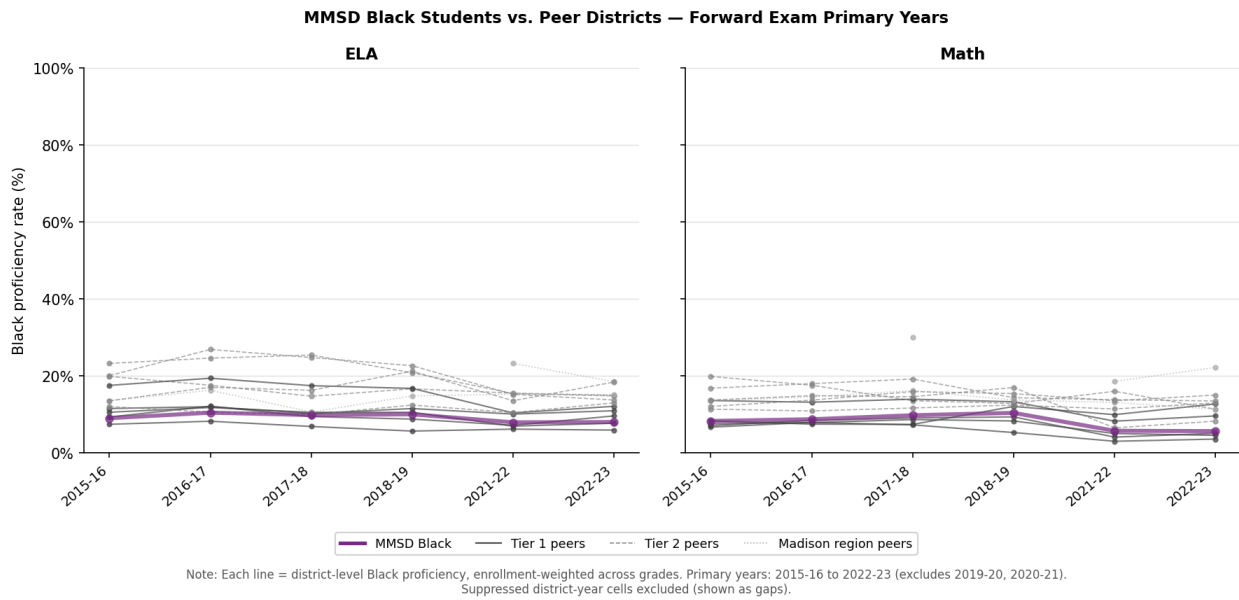


Figure 9: MMSD Black Students vs. Peer District Black Students

MMSD Black students achieve ELA proficiency rates of approximately 8–10% across primary years. This is substantially below Milwaukee Black students (approximately 16–22%) and below several Tier 2 peer districts as well. The same pattern holds in Mathematics. Figure 10 shows a similar pattern for Hispanic students: MMSD Hispanic proficiency (~15–17%) is broadly comparable to or below peer districts.

These findings imply that the large within-MMSD racial gap is not primarily the result of minority students performing unusually well in MMSD — rather, it reflects the unusually high performance of MMSD White students combined with minority students performing no better (and in some dimensions, worse) than in peer districts.

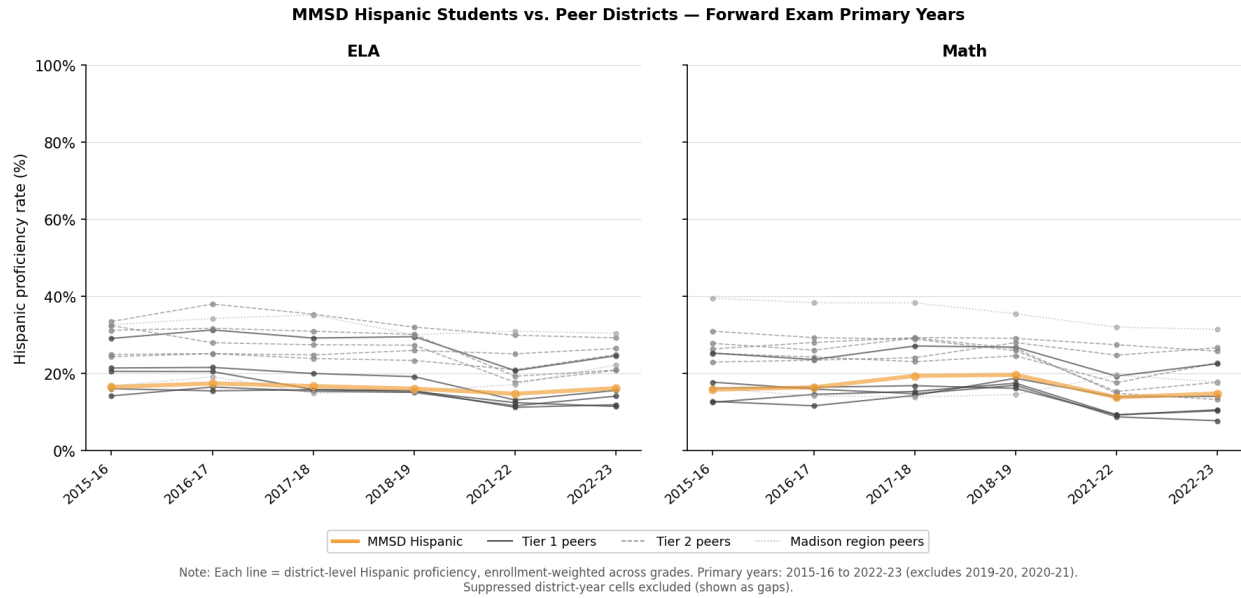


Figure 10: MMSD Hispanic Students vs. Peer District Hispanic Students

MMSD Minority vs. Non-MMSD Wisconsin White

To further underscore the outlier-SES status of MMSD White students, Figure 11 compares MMSD Black and Hispanic proficiency directly to the proficiency of White students in all Wisconsin districts except MMSD.

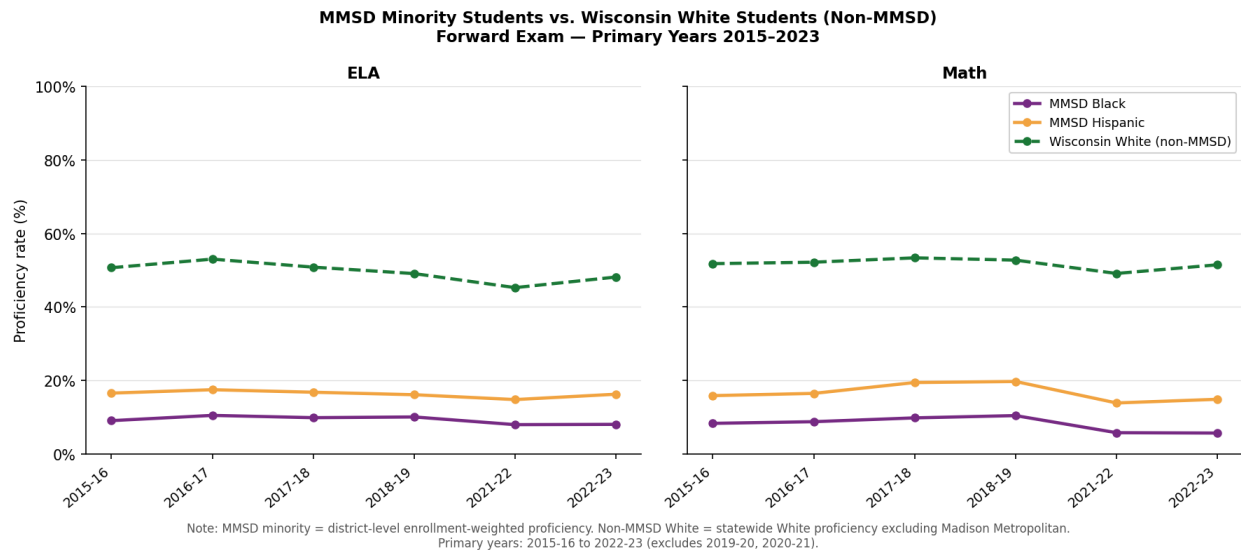


Figure 11: MMSD Minority Students vs. Non-MMSD Wisconsin White Students

Non-MMSD Wisconsin White students achieve ELA proficiency rates of approximately 45–53%, while MMSD Black students achieve only 8–10% and MMSD Hispanic students

achieve 15–17%. The gap between MMSD Black students and non-MMSD Wisconsin White students is approximately 40–43 percentage points — comparable to the statewide Black-White gap reported in Section . This comparison illustrates that MMSD minority students are performing substantially below even the statewide White average, not merely below the exceptional MMSD White population.

School-Level White and Minority Proficiency (Statewide)

At the school level, DPI suppresses race-specific proficiency for small subgroups, so only a subset of buildings have both White and Black (or White and Hispanic) proficiency rates observed. Figure 12 summarizes Wisconsin public schools with non-suppressed data: Panel A places White proficiency on the horizontal axis and Black proficiency on the vertical axis; Panel B uses Hispanic proficiency on the vertical axis. Madison Metropolitan, Milwaukee, and all other districts are shown in distinct colors; the figure note reports Pearson correlations between the axes for the overall sample and for each of those three district groups.

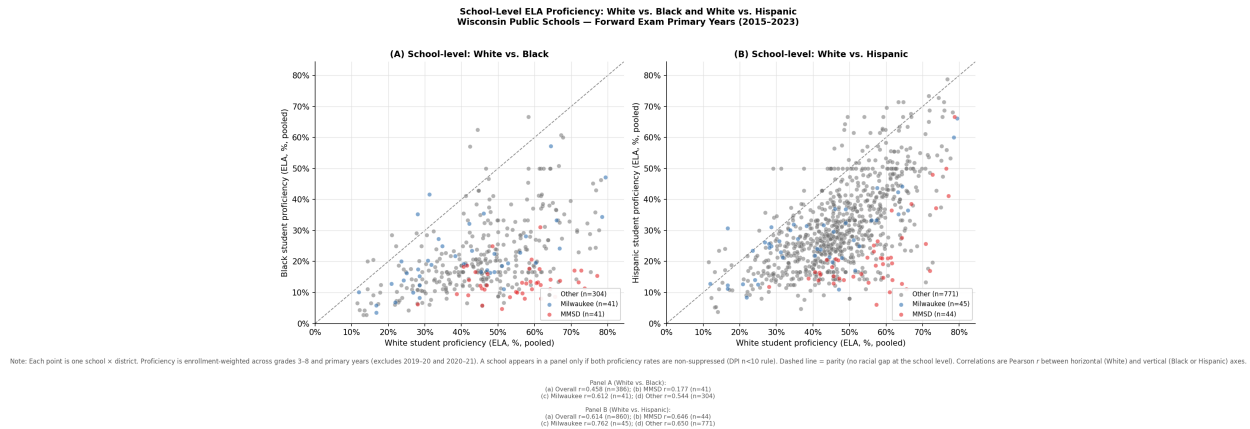


Figure 12: School-Level White vs. Black and White vs. Hispanic ELA Proficiency

Two patterns stand out. First, Black student proficiency in MMSD is only weakly associated with the White proficiency level of the same school: among MMSD schools with both rates observed, the correlation is modest (approximately 0.2), whereas statewide and in the “other districts” remainder the White–Black correlation is substantially higher (roughly 0.45–0.55 in those subsamples, and higher still in Milwaukee). In practical terms, attending an MMSD school with stronger White proficiency is not a strong signal of stronger Black proficiency in that building—consistent with the view that the within-MMSD racial gap is not primarily driven by a simple “school quality” ordering on which Black and White students move in tandem. Second, the White–Hispanic correlation is stronger and more consistent across subsamples: overall and within MMSD, Milwaukee, and other districts, correlations are

typically in the 0.6–0.8 range (see figure note for exact values). Hispanic proficiency co-moves more tightly with White proficiency at the school level than Black proficiency does, especially in MMSD. These school-level scatters complement the within-MMSD boxplots and decomposition below: they describe alignment of racial group means within schools, not causal mechanisms.

Within-MMSD School Distribution

Figure 13 shows the distribution of school-level proficiency within MMSD by race and year, pooled across grade and subject.

Analytic sample note. The Forward Exam covers grades 3–8, which spans both elementary schools (grades 3–5) and middle schools or K–8 schools (grades 6–8) in MMSD. The MMSD boundary review specifically targets elementary school attendance zones, but we include all school levels in the decomposition to maximize statistical power and to characterize the full picture of within- vs. between-school sorting. Of MMSD’s approximately 51 elementary and K–8 schools, between 27 and 36 schools have sufficient non-suppressed data for both Black and White students in a given year (see Appendix A for suppression details). The schools excluded from the decomposition are predominantly smaller schools where one or both race groups fall below the DPI reporting threshold of 10 students; they are not systematically different in overall proficiency. Readers interested specifically in the elementary-school context should note that the between-school share of the gap would likely differ somewhat if the analysis were restricted to grades 3–5 alone.

Note: Each box represents the distribution of school-level proficiency rates for that race group across MMSD schools, for non-suppressed schools only. School-level rates are enrollment-weighted averages across all non-suppressed grade $\times$ subject cells. The analytic sample covers 27–36 schools per year (of MMSD’s ~51 elementary and K–8 buildings).

Several features of this figure are noteworthy. First, the level of Black proficiency in MMSD schools is uniformly low — the median MMSD school has Black proficiency of approximately 5–12%, and there is no MMSD school where Black students approach the White proficiency rate. Second, the within-school decomposition reveals that approximately 82% of the MMSD BW gap is within individual schools, and only 18% is between schools.¹

¹The school-level proficiency rate used in Figure Figure 13 and the decomposition pools ELA and Mathematics scores into a single enrollment-weighted average per school–race–year cell. We verify that this pooling does not affect the main result: running the decomposition separately for ELA only and Mathematics only yields within-school shares of 82.2% and 82.2%, respectively — indistinguishable from the pooled estimate of 82.1%. The between-school share ranges from 15.6% to 19.1% across individual years and subjects, with no systematic difference between ELA and Mathematics. Full subject-specific results are reported in

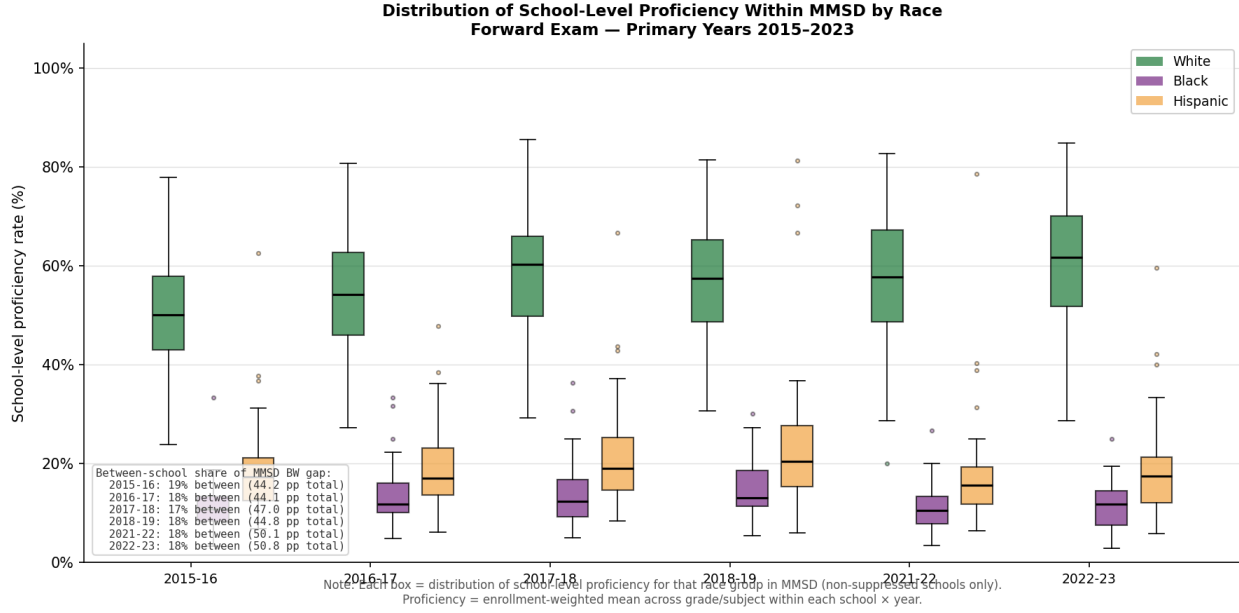


Figure 13: Distribution of School-Level Proficiency Within MMSD by Race

This decomposition is the central descriptive finding for the policy context. The MMSD attendance boundary review could, in principle, redistribute students across schools and thereby affect the between-school component of the gap. The full between-school component — approximately 18% of a roughly 47 pp gap, or about 8–9 percentage points — represents the upper bound on gap reduction achievable through complete equalization of racial composition across schools. In practice, the achievable reduction would be smaller: Gillani et al. (2023) found that optimal algorithmic redesign achieved a median 14% reduction in segregation (not the achievement gap directly) across 98 US elementary districts, and even that required approximately 20% of students to change schools. The pathway from reduced school segregation to reduced achievement gaps is not direct, and the actual impact of MMSD’s boundary review will depend on how instructional resources, course placement, peer effects, and school climate respond to reassignment — factors outside the scope of this analysis.

Conclusion

Wisconsin’s racial achievement gap is large, national in rank, and persistent over time. Our analysis documents a statewide Black-White proficiency gap of approximately 38–44 pp on the Forward Exam (depending on subject), with roughly half attributable to between-district

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sorting and half within districts.

For MMSD specifically, three findings stand out. First, MMSD Black and Hispanic students perform no better than — and in some cases worse than — same-race students in Milwaukee and other peer districts, despite MMSD’s higher overall district ranking. This reflects the outlier-SES composition of MMSD White students, not unusually good outcomes for minority students. Second, the gap between MMSD minority students and non-MMSD Wisconsin White students (approximately 40 pp in ELA) is nearly as large as the statewide average BW gap, underlining that MMSD minority students are struggling relative to a representative White comparison group, not just relative to the extraordinary MMSD White population. Third, and most directly relevant to the boundary review, 82% of the MMSD BW gap exists within individual schools. The between-school component — approximately 8–9 percentage points — is the portion most directly targeted by student reassignment; the remaining within-school 82% reflects factors that boundary changes cannot directly alter.

These findings do not imply that boundary changes are without value — school integration has other benefits beyond direct test score effects, including exposure to diverse peers and reduced concentration of poverty in individual schools (Reardon et al. 2024). But the decomposition suggests that boundary redesign, even under optimistic assumptions, would address at most the between-school share of the gap. The actual achievable reduction will depend on whether and how instructional quality, course placement, and school climate change in response to reassignment. Policies targeting the within-school component of the gap — instructional quality, early intervention, teacher distribution, and the school poverty concentration that underlies the gap — would need to be central to any comprehensive equity strategy.

Finally, a supplemental analysis using the prior Wisconsin Knowledge and Concepts Examination (WKCE) scale-score data (2008–09 to 2013–14, presented in Appendix D) confirms that the patterns documented above are not artifacts of the Forward Exam era. Statewide Black-White scale-score gaps of 44–55 points were persistent across the WKCE era, and approximately 55–58% was attributable to between-district sorting — broadly consistent with the Forward Exam decomposition. MMSD Black students also performed comparably to Milwaukee Black students on the WKCE, reinforcing the conclusion that MMSD’s large overall racial gap is driven primarily by its exceptional White student population rather than by above-average outcomes for its minority students.

Data Availability

All analysis code, data download scripts, and figure-generation scripts are available at [GitHub repository link]. Raw data are from Wisconsin DPI and are publicly available via WISEdash.

Acknowledgments

During the preparation of this work the author used Cursor 2.6.22 in order to assist in code editing, text edits, and reference management. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for its content. The AI-assisted workflow used in this project is built on the academic template developed by Sant’Anna (2025).

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Appendix

A. Suppression Rates by Race and Level

Suppression occurs when a student subgroup has fewer than ten tested individuals in a given school $\times$ grade $\times$ subject $\times$ year cell. Rather than imputing suppressed cells, this study treats them as missing throughout. The tables below document the extent of suppression in the Forward Exam panel (primary years 2015–16 to 2022–23, excluding 2019–20 and 2020–21).

White students have the lowest suppression rates (36% at both levels) because they constitute the majority group in virtually every district and school. Black students, despite being the primary minority group of interest, have the second-highest suppression rate (82%) because Black enrollment is geographically concentrated in a small number of districts and is absent or very small in the majority of Wisconsin’s 421 school districts. American Indian students have the highest suppression rate (91–94%) for the same reason. The within-between decomposition and peer district analyses are restricted to cells where both groups have valid

proficiency data, so the effective sample for Black-White decompositions is 30–45 districts per year.

Suppression rates are stable across years within each race group, which confirms that the patterns documented in the main body are not driven by year-specific changes in the tested population.

B. Decomposition Results — Full Table

The within-between decomposition is computed separately for each year $\times$ subject $\times$ grade $\times$ gap-pair cell at the district level. `?@tbl-decomp-full` presents grade-averaged results (unweighted across grades 3–8) for each year and subject, for the Black-White and Hispanic-White gap pairs. The full grade-by-grade table is available in `output/tables/decomposition_results.csv`.

B.2 Within-MMSD School Decomposition by Subject

`?@tbl-decomp-subject` reports the school-level within-between decomposition for MMSD run separately for ELA, Mathematics, and the pooled specification. This table supports the footnote in Section showing that the 82% within-school share is robust to the choice of subject.

Across all six primary years, the between-school share for the BW gap ranges from 15.6% to 19.1% regardless of whether ELA, Mathematics, or both subjects are used. The within-school share is consistently 80–84%. Post-COVID years (2021–22, 2022–23) show somewhat larger total gaps (T 49–52 pp vs. 43–47 pp pre-COVID), consistent with differential learning recovery by race and school type.

C. COVID Sensitivity Check

The primary analysis excludes 2019–20 (no assessment administered) and 2020–21 (COVID disruption year with partial in-person schooling, depressed overall proficiency rates, and altered testing conditions). This appendix presents key results with 2020–21 included to assess the sensitivity of the main findings.

Key findings:

- In 2020–21, overall proficiency rates declined sharply for all groups. White proficiency fell from a pre-COVID average of 51% to 42%; Black proficiency fell from 12% to 5%; Hispanic proficiency fell from 24% to 15%.
- The Black-White gap in 2020–21 was 37.6 percentage points — slightly narrower than the pre-COVID average of 40 pp. This reflects the fact that COVID-related learning loss, while severe in absolute terms for all groups, was somewhat larger for White students in proficiency rate terms (a ceiling effect may play a role: White students had further to fall above the proficiency threshold).
- By 2022–23 the gap had returned to 41.4 pp, statistically indistinguishable from the 2015–2019 average.
- Crucially, including 2020–21 in the trend does not alter any of the paper’s core conclusions. The gap was large and stable before, during, and after the COVID disruption. The decomposition finding (47% between, 53% within) is also robust: the between-district share in 2020–21 (not shown, computed separately) is within 2 pp of the primary-year average.
- Because 2020–21 assessment conditions were materially different from all other years (reduced participation, non-standard testing modalities, and uncertainty about comparability of scores), we exclude it from the primary analysis to avoid confounding the trend narrative with a one-year disruption. All qualitative conclusions are robust to its inclusion.

D. WKCE-Era Supplemental Analysis (2008–09 to 2013–14)

Overview

This appendix presents parallel analyses to the main body of the paper, using the Wisconsin Knowledge and Concepts Examination (WKCE) scale-score data for the school years 2008–09 through 2013–14.

Why a separate era? The WKCE and the Forward Exam are distinct assessment instruments with different items, scales, and proficiency thresholds. They cannot be bridged or placed on a common scale. We therefore treat them as entirely separate analytical windows: the Forward Exam era (2015–2023) constitutes the main analysis, while this appendix provides a historical cross-check using the WKCE era.

What the WKCE measures. The WKCE uses a vertically scaled developmental score in each subject. Unlike proficiency rates, scale scores are designed to be comparable across grade

levels and across years within the same assessment program. For the purposes of comparing group means, scale scores have the advantage of preserving the full distribution rather than collapsing it to a binary above/below threshold. The substantive interpretation is that a gap of, say, 45 scale-score points represents the difference in average developmental standing between White and Black students.

Analysis window. We restrict to school years 2008–09 through 2013–14. The year 2006–07 produced no valid data in the processed panel. The year 2007–08 has valid scale scores but entirely missing enrollment denominators, precluding enrollment-weighted aggregation. The year 2013–14 is the final WKCE administration; the Forward Exam debuted in 2014–15.

Subjects and grades. We focus on Reading and Mathematics, grades 3–8, to mirror the main Forward Exam analysis.

Suppression. WKCE files report a district-level scale score as missing (“suppressed”) when the student subgroup had fewer than approximately 10 students tested in that grade. After aggregating across grades 3–8 within each district–race–year–subject cell, the district-level suppression rate for Black students is approximately 35% and for Hispanic students approximately 45%. The within-between decomposition is therefore based on the subset of districts (58–79 for Black-White cells) where both racial groups have valid data.

D.1 Statewide Racial Scale-Score Gaps

Figure 14 shows the enrollment-weighted statewide gap between White and minority students in each year of the WKCE era, for Reading and Mathematics.

Key findings:

- The Black-White gap in Reading is 44–48 scale-score points and is remarkably stable over 2008–2014, with no consistent upward or downward trend.
- The Black-White gap in Mathematics is 49–55 scale-score points and exhibits a modest widening trend over the WKCE era, reaching 55 points in 2013–14.
- The Hispanic-White gap is smaller than the Black-White gap in both subjects: approximately 34–37 points in Reading and 34–36 points in Mathematics, consistent with the pattern in the Forward Exam era.
- White statewide mean scale scores are stable around 504 in Reading and 512 in Mathematics, while Black mean scores are stable around 457–461 in Reading. This stability across six years indicates the gap is structural, not driven by year-to-year

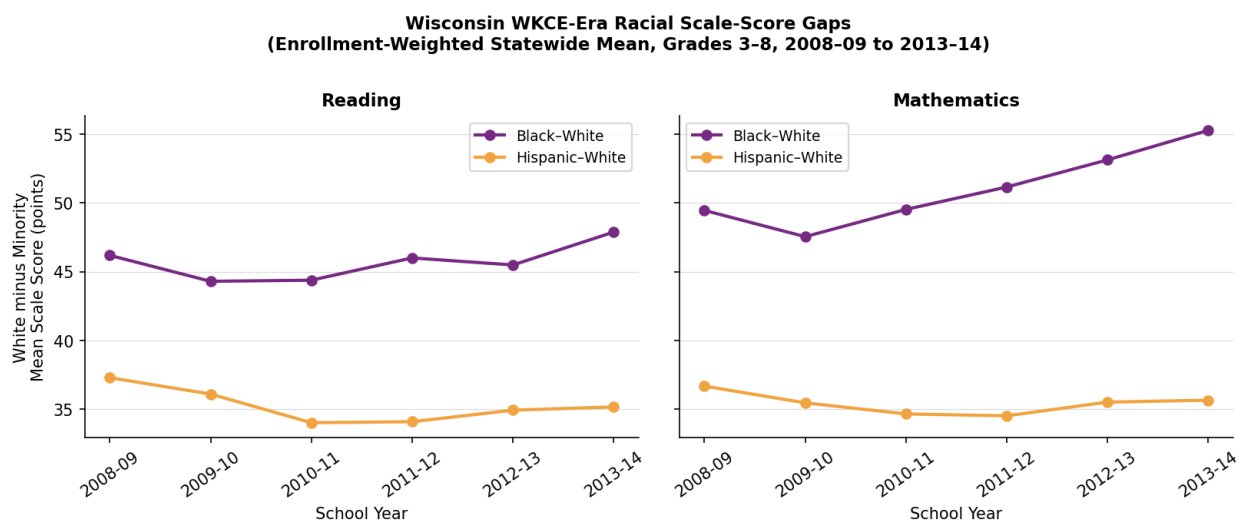


Figure 14: Statewide WKCE Racial Scale-Score Gaps by Year (Grades 3–8)

fluctuations.

D.2 Scale-Score Gaps by Grade

Figure 15 shows average BW and HW gaps by grade (pooled across all WKCE years).

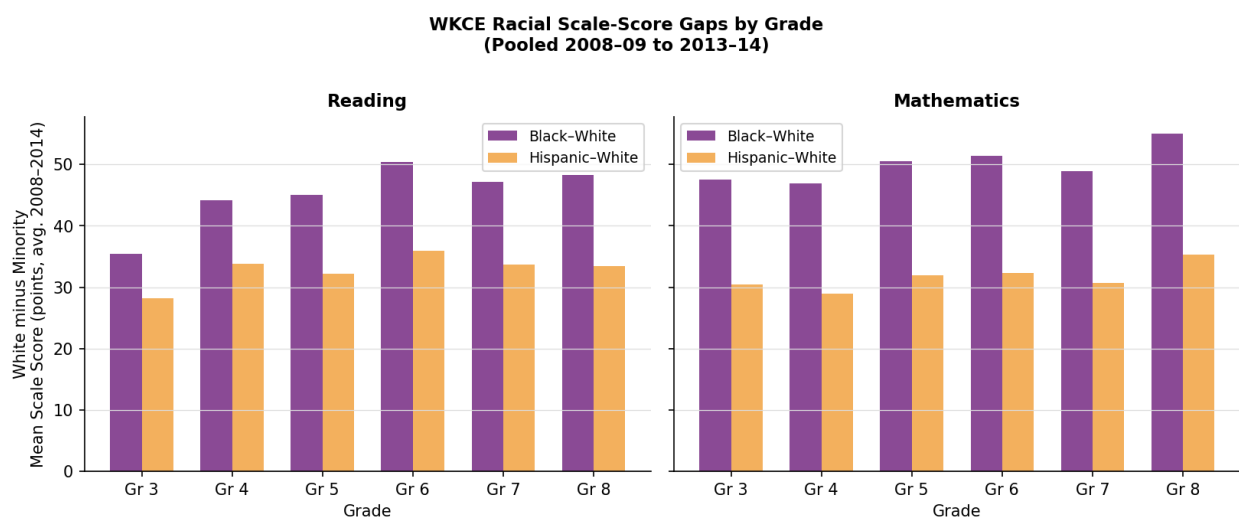


Figure 15: WKCE Racial Scale-Score Gaps by Grade (Pooled 2008–09 to 2013–14)

The grade gradient in the WKCE era shows little systematic widening with grade in Reading, with BW gaps ranging from roughly 43–48 points across grades 3–8. In Mathematics, there is evidence of a slight upward gradient, with gaps somewhat larger in grades 6–8 than in grades

3–5. This pattern — larger math gaps at higher grades — is consistent with cumulative disadvantage or differential access to advanced coursework, and echoes similar patterns in national data (Reardon et al. 2019).

D.3 Within-Between Decomposition

We apply the same $T = B + W$ decomposition as in Section of the main text, replacing proficiency rates with mean scale scores. The decomposition is performed at the district level, aggregating grades 3–8 within each district-race-year-subject cell.

Figure 16 shows the within and between shares of the BW and HW gap for each year and subject. The total gap T (in scale-score points) is printed above each bar.

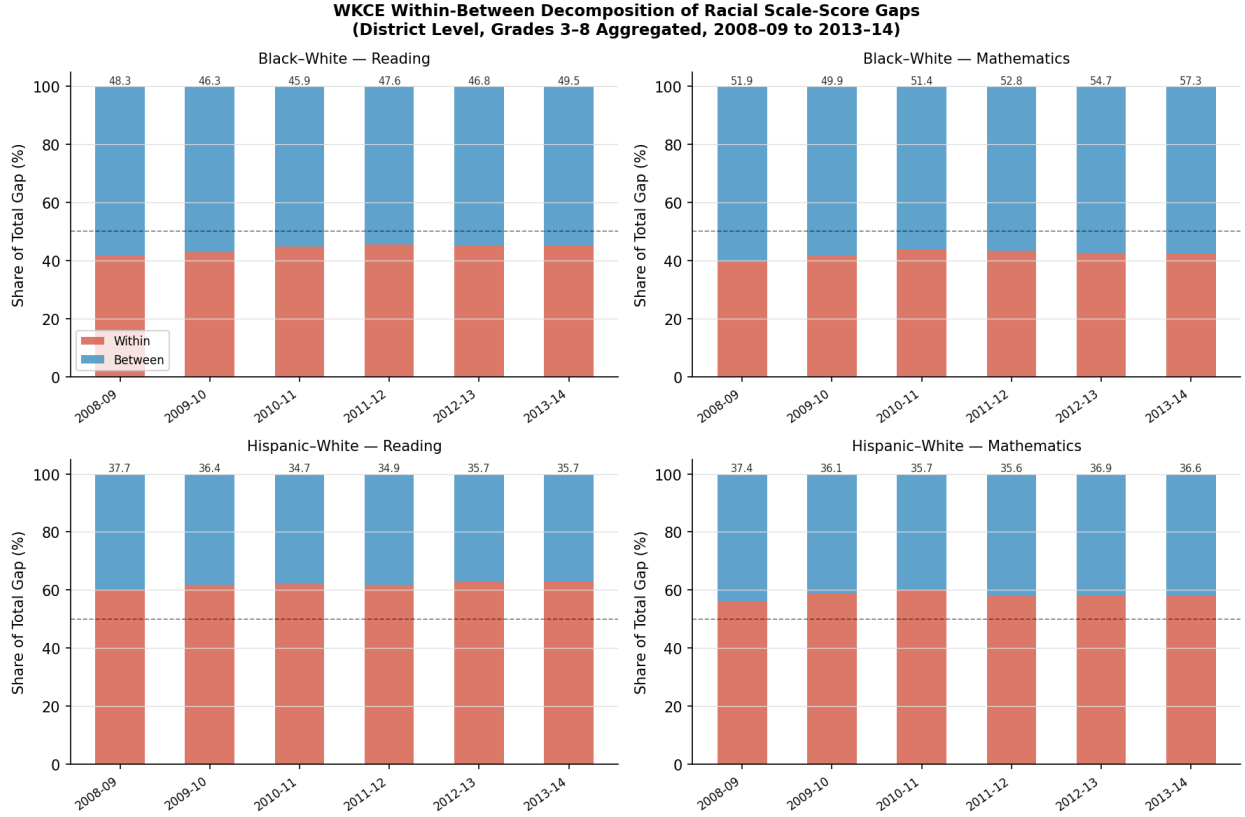


Figure 16: WKCE Within-Between Decomposition of Racial Scale-Score Gaps

Key findings from the WKCE-era decomposition:

- For the Black-White gap, 55–58% is attributable to between-district sorting in Reading, and 57–60% in Mathematics. This is somewhat higher than the Forward Exam era estimate of ~47%, potentially reflecting greater residential segregation in the earlier period or differences in the set of districts included (the WKCE decomposition includes only the smaller set of districts with valid data for both races).
 - The between share of the BW gap shows a modest declining trend over the WKCE era, from 58% in 2008–09 to 55% in 2013–14 (Reading). This suggests a slight increase in within-district sorting or integration at the district level over this period.
 - For the Hispanic-White gap, the between share is substantially lower: only 37–40% in Reading and similar in Mathematics. The majority of the HW gap is therefore within districts — consistent with the Forward Exam era and with the broader pattern that Hispanic students in Wisconsin are more geographically integrated across districts than Black students.
-

D.4 MMSD vs. Peer Districts (WKCE Era)

Figure 17 compares MMSD Black and Hispanic mean scale scores in Reading and Mathematics to those of the same peer districts used in the main analysis.

Key findings:

- MMSD Black students (mean Reading score ~453) perform almost identically to Milwaukee Black students (~451) across the entire WKCE era. This directly mirrors the Forward Exam finding and is particularly striking given that Milwaukee serves roughly 7× as many Black students as MMSD.
 - MMSD Hispanic students score slightly above several Tier 1 peers (Milwaukee Hispanic, Racine Hispanic) in Reading, though the differences are modest (3–6 scale-score points).
 - Tier 1 urban districts (Milwaukee, Racine, Kenosha) consistently cluster together at the lower end of the distribution, while Madison-region peers (Verona, Middleton) and some mid-size districts (Waukesha, Janesville) show higher minority mean scores.
 - These patterns provide historical confirmation that MMSD’s large overall racial gap in the Forward Exam era is not a new phenomenon — it was equally present in the WKCE era.
-

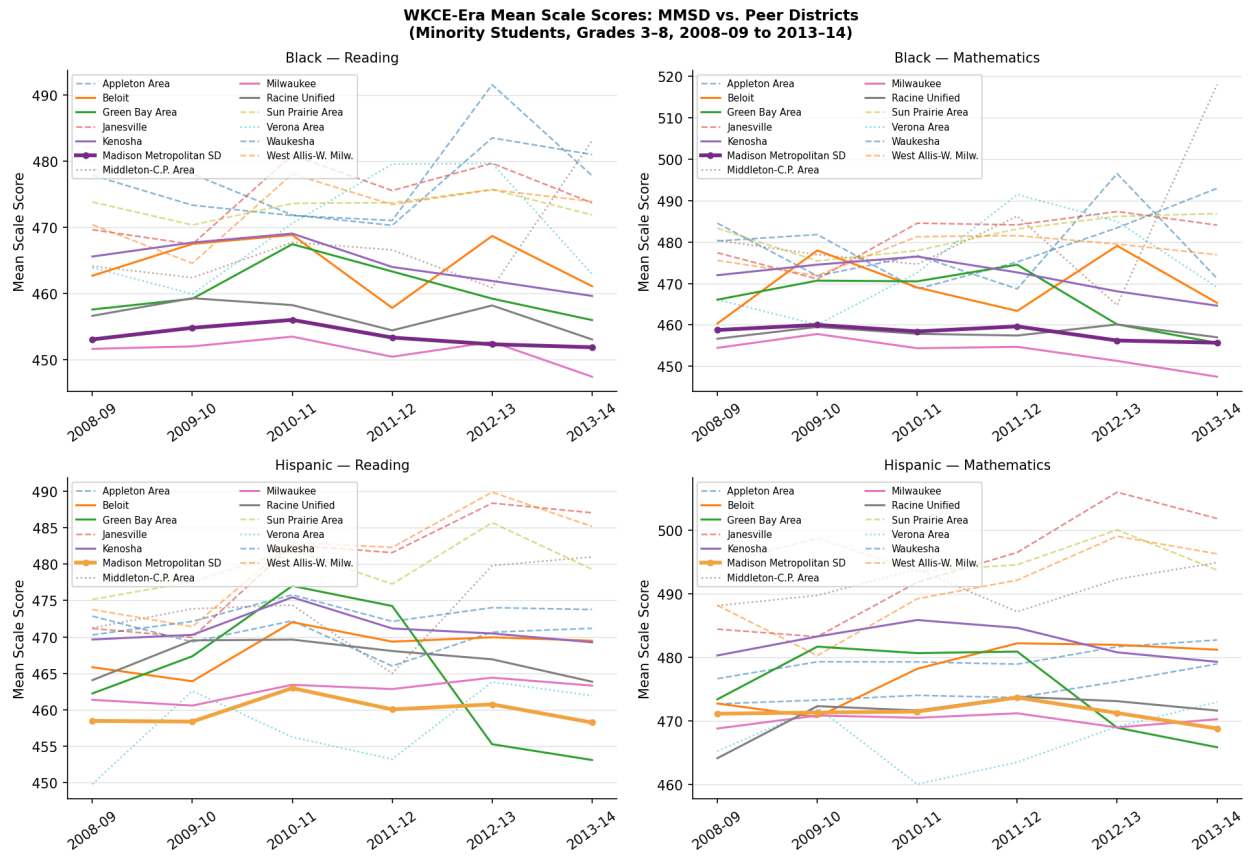


Figure 17: WKCE-Era Mean Scale Scores: MMSD vs. Peer Districts

D.5 MMSD Minority vs. Non-MMSD White (WKCE Era)

Figure 18 extends the peer comparison in Section by comparing MMSD minority students directly to non-MMSD White students statewide.

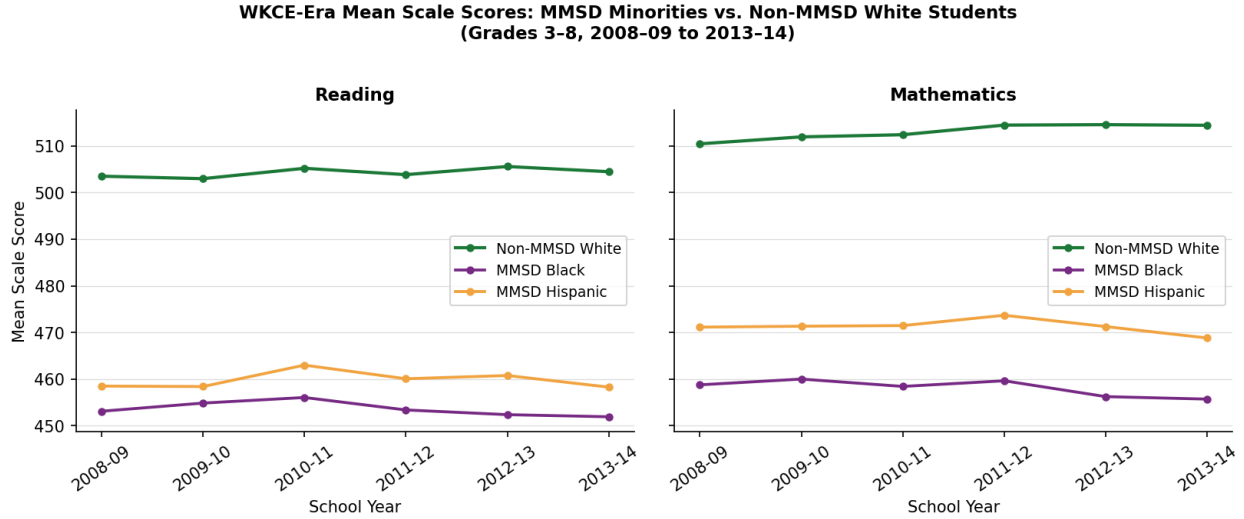


Figure 18: WKCE-Era Mean Scale Scores: MMSD Minorities vs. Non-MMSD Wisconsin White Students

Non-MMSD White students average approximately 504 scale-score points in Reading and 512–515 in Mathematics, stable across all WKCE years. MMSD Black students average approximately 453 points in Reading — a gap of roughly 50–51 points from the non-MMSD White average. MMSD Hispanic students average approximately 459–463 points in Reading — a gap of roughly 41–45 points.

These WKCE-era results are closely analogous to the Forward Exam finding that MMSD Black students score approximately 40–43 percentage points below the non-MMSD White proficiency rate. Both measures confirm that MMSD minority students are substantially below the statewide White benchmark, and that this gap has been persistent for at least a decade before the current Forward Exam era began.

D.5b Grade 10 Profile (WKCE Era)

The WKCE tested grade 10 students in Reading, Mathematics, and three other subjects. Although grade-10 district-level suppression is higher (~83% at the district $\times$ race level), state-level aggregation produces stable estimates for 2008–09 through 2012–13 (note: grade-10 data in 2013–14 are incomplete and excluded).

Figure 19 compares the average BW and HW scale-score gap at grades 3–8 to that at grade 10, for Reading and Mathematics.

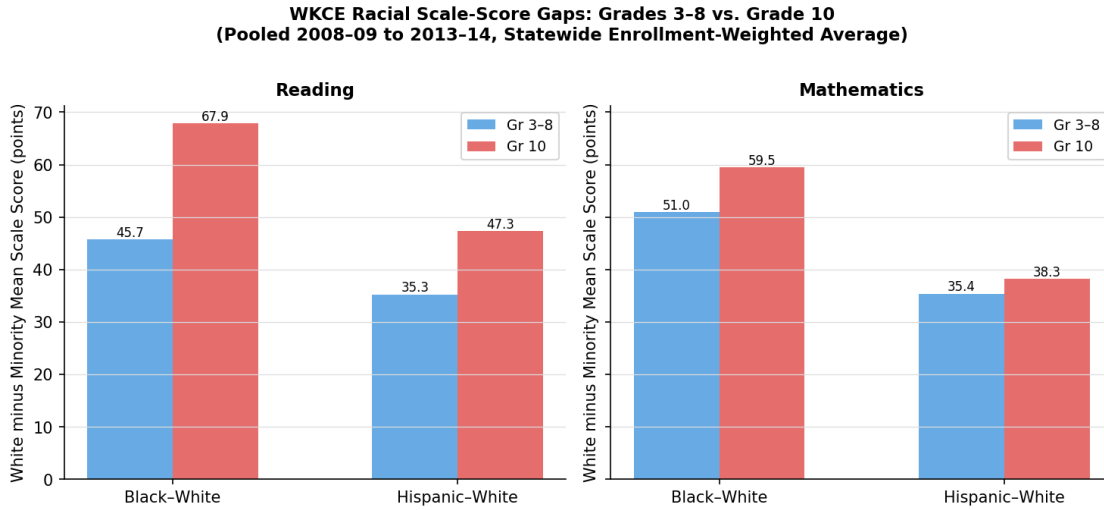


Figure 19: WKCE Scale-Score Gaps: Grades 3–8 vs. Grade 10

The grade-10 BW gap in Reading is 64–74 scale-score points, compared to 44–48 points in grades 3–8. The same pattern holds in Mathematics. This widening is consistent with cumulative disadvantage: the gap grows over the school career. However, care is needed in interpretation — grade-10 scale scores are on a higher part of the WKCE developmental scale, so a larger absolute gap does not necessarily imply a larger standardized gap. Expressed in within-grade standard deviations, the grade-10 gap is roughly comparable to or modestly larger than the grades 3–8 gap, consistent with the ACT evidence from the Forward Exam era showing no convergence through grade 11.

D.6 Cross-Era Comparison Summary

The cross-era comparison reveals striking consistency in the structure of Wisconsin’s racial achievement gap across two entirely different assessment instruments and two separate time periods spanning approximately 15 years. The between-district share of the BW gap is somewhat higher in the WKCE era (55–60%) than the Forward era (~45–47%), though this difference may partly reflect the smaller number of districts included in the WKCE decomposition. The conclusion that a substantial portion of the Black-White gap is between districts — and therefore cannot be addressed by within-district school boundary changes — is robust across both eras.

The MMSD-specific finding is also confirmed across eras: MMSD Black and Hispanic students perform at approximately the same level as their peers in Milwaukee and other urban districts. MMSD’s extraordinary overall reputation and its extraordinarily high-SES White population generate a large racial gap in test scores, but this gap does not reflect above-average outcomes for MMSD minority students. This finding holds in 2008 and in 2023.

E. Economic Disadvantage vs. Non-Disadvantage (School-Level)

The main text uses race/ethnicity cuts from the Forward Exam. As a supplementary exercise, we replicate the school-level scatter design as closely as the public data allow using DPI’s Economic Status breakdown: “Not Econ Disadv” (not economically disadvantaged) and “Econ Disadv” (economically disadvantaged). This is not a substitute for the Black–White or Hispanic–White analysis: economic status is correlated with race, but it is a distinct construct, and we do not observe race $\times$ poverty cells at suppressed-safe school-level sample sizes here.

Figure 20 plots, for each school with both rates non-suppressed, non-ED proficiency on the horizontal axis and ED proficiency on the vertical axis (ELA only, pooled across grades 3–8 and primary years, same construction as Figure 12). Colors distinguish MMSD, Milwaukee, and other districts. Pearson correlations by subsample are printed in the figure note.

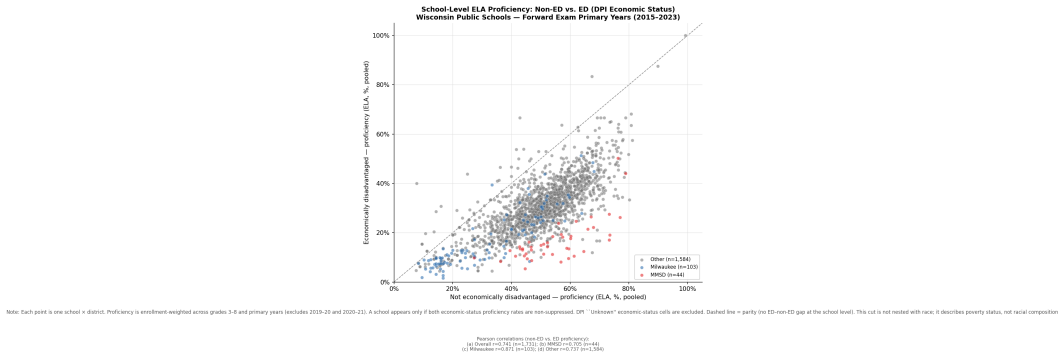


Figure 20: School-Level Non-ED vs. ED ELA Proficiency (DPI Economic Status)

Pattern. The association between non-ED and ED proficiency at the same school is strong and positive statewide (correlation approximately 0.74 in the overall sample), and similarly high within MMSD, Milwaukee, and other districts—substantially stronger than the White–Black school-level correlation in MMSD in Figure 12. In other words, schools that are better for non-economically disadvantaged students also tend to be better for economically disadvantaged students in a way that Black and White proficiency do not move together

within MMSD buildings. This appendix is descriptive only; it does not identify causal effects of economic status or school assignment.